

Abalone Age Prediction - Complete Feature Reference

Summary Statistics

Category	Count
Original Features	10
Ratio Features	7
Volume/Density	4
Geometric Means	4
Binary Flags	5
Polynomial Terms	5
Interaction Terms	15
Log Transforms	7
Sqrt Transforms	3
Spline Terms	3
Total Unique Features	63

1. ORIGINAL FEATURES (10)

Feature	Type	Units	Description
id	Integer	-	Unique identifier
Sex	Categorical	M/F/I	Male, Female, or Infant
Length	Numeric	Inches	Longest shell measurement
Diameter	Numeric	Inches	Perpendicular to length
Height	Numeric	Inches	Height with meat inside

Feature	Type	Units	Description
Weight	Numeric	Grams	Whole abalone weight
Shucked Weight	Numeric	Grams	Meat weight
Viscera Weight	Numeric	Grams	Gut weight
Shell Weight	Numeric	Grams	Dried shell weight
Age	Integer	Years	Target variable (Rings + 1.5)

Models Using Original Features Only: M1 (Baseline)

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2. RATIO FEATURES (7)

Feature	Equation	Description	Models
shucked_ratio	Shucked Weight / Weight	Meat proportion	M3-M9, M11-M15, M19-M35
viscera_ratio	Viscera Weight / Weight	Gut proportion	M11, M16-M18, M36-M38
shell_ratio	Shell Weight / Weight	Shell proportion	M3-M9, M11-M15, M19-M35
meat_ratio	(Shucked + Viscera) / Weight	Soft tissue proportion	Available
height_diameter_ratio	Height / Diameter	Flatness indicator	M11, M16-M18, M36-M38
diameter_length_ratio	Diameter / Length	Roundness indicator	Available
height_length_ratio	Height / Length	Height proportion	Available

Key Insight: shell_ratio and shucked_ratio are among the most predictive ratios.

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3. VOLUME & DENSITY FEATURES (4)

Feature	Equation	Description	Models
volume_ellipsoid	$\frac{(4/3) * \pi * (L/2) * (D/2) * (H/2)}{}$	3D size as ellipsoid	M11, M16-M18, M36-M38

Feature	Equation	Description	Models
volume_cylinder	$\pi * (D/2)^2 * H$	Alternative volume	Available
density	Weight / volume_ellipsoid	Compactness measure	M10-M11, M14, M16-M18, M28, M33, M36-M38
surface_area	$\frac{4\pi * ((L*D)^{1.6075} / 3)^{(1/1.6075)}}$	Outer shell area	Used for shell_thickness_proxy

Key Insight: density captures how "compact" an abalone is for its size.

4. GEOMETRIC MEAN FEATURES (4)

Feature	Equation	Description	Models
size_mean	$(Length * Diameter * Height)^{(1/3)}$	Overall 3D size	M2-M15, M19-M35
size_norm	$\sqrt{L^2 + D^2 + H^2}$	Euclidean norm of dimensions	Available
weight_mean	$(Shucked * Viscera * Shell)^{(1/3)}$	Combined weight measure	Available
weight_norm	$\sqrt{Shucked^2 + Viscera^2 + Shell^2}$	Euclidean norm of weights	Available

Key Insight: size_mean is the single most important engineered feature.

5. DERIVED FEATURES (2)

Feature	Equation	Description	Models
shell_thickness_proxy	Shell Weight / surface_area	Shell development indicator	M2-M15, M19-M35
bio_flag	Case-based	Biological anomaly flag	Used to create binary flags

6. BINARY FLAG FEATURES (5)

Feature	Equation	Description	Models
is_infant	if_else(Sex == "I", 1, 0)	Infant indicator	M2-M15, M19-M38
is_male	if_else(Sex == "M", 1, 0)	Male indicator	Available
is_female	if_else(Sex == "F", 1, 0)	Female indicator	Available
is_slow_growth	if_else(bio_flag == "Infant_too_old", 1, 0)	Infant > 6 years old	M4-M9, M11-M15, M19-M38
is_early_mature	if_else(bio_flag == "Adult_too_young", 1, 0)	Adult < 5 years old	Available

Key Insight: is_infant is critical - infants have different growth patterns.

7. POLYNOMIAL FEATURES (5)

Feature	Equation	Description	Models
size_mean_squared	size_mean ²	Quadratic size term	M5-M8, M11-M15, M24, M26, M28-M33, M35-M38
size_mean_cubed	size_mean ³	Cubic size term	M13, M26, M32, M36-M38
size_mean_quartic	size_mean ⁴	Quartic size term	M26
weight_squared	Weight ²	Quadratic weight term	Available
shell_ratio_squared	shell_ratio ²	Quadratic shell ratio	Available

Key Insight: size_mean_squared significantly improves model fit (captures diminishing returns).

8. INTERACTION FEATURES (15)

Feature	Equation	Description	Models
size_shell_interaction	size_mean * shell_ratio	Size × shell proportion	M12, M15, M28-M35
size_shucked_interaction	size_mean * shucked_ratio	Size × meat proportion	M15, M29, M33-M35, M36-M38
size viscera_interaction	size_mean * viscera_ratio	Size × gut proportion	M36-M38
weight_infant_interaction	Weight * is_infant	Weight effect for infants	M12, M15, M25, M28-M35
shell_infant_interaction	shell_ratio * is_infant	Shell effect for infants	M15, M28-M35
size_weight_interaction	size_mean * Weight	Size × weight	M6
shell_shucked_interaction	Shell Weight * Shucked Weight	Shell × meat weight	Available
density_size_interaction	density * size_mean	Density × size	M14, M33
weight_shell_interaction	Weight * shell_ratio	Weight × shell proportion	M29, M36-M38
weight_shucked_interaction	Weight * shucked_ratio	Weight × meat proportion	M29
shell_shucked_ratio_interaction	shell_ratio * shucked_ratio	Shell × meat ratios	Available
infant_size_squared	is_infant * size_mean ²	Quadratic size for infants	M31, M33, M36-M38
slow_growth_size	is_slow_growth * size_mean	Size for slow growers	M30, M33, M36-M38
density_infant_interaction	density * is_infant	Density for infants	M28, M33, M36-M38
thickness_infant_interaction	shell_thickness_proxy * is_infant	Thickness for infants	Available

Key Insight: Interactions are the biggest performance boosters. M15 (with 4 interactions) is consistently top-performing.

9. LOG TRANSFORM FEATURES (7)

Feature	Equation	Description	Models
$\log_size_mean$	$\log(size_mean)$	Linearizes size-age	M9, M20
$\log_weight$	$\log(Weight)$	Linearizes weight-age	M19, M20, M25, M35
$\log_shell_weight$	$\log(Shell\ Weight)$	Linearizes shell-age	M19, M20, M25
$\log_shucked_weight$	$\log(Shucked\ Weight)$	Linearizes meat-age	M20
$\log_viscera_weight$	$\log(Viscera\ Weight)$	Linearizes gut-age	M20
$\log_density$	$\log(density)$	Linearizes density	Available
$\log_shell_thickness$	$\log(shell_thickness_proxy)$	Linearizes thickness	Available

Key Insight: Log transforms help address heteroscedasticity in residuals.

10. SQRT TRANSFORM FEATURES (3)

Feature	Equation	Description	Models
$\sqrt{size_mean}$	$\sqrt{size_mean}$	Alternative to log	M27
$\sqrt{weight}$	$\sqrt{Weight}$	Alternative to log	M27
$\sqrt{shell_weight}$	$\sqrt{Shell\ Weight}$	Alternative to log	M27

11. SPLINE FEATURES (3)

Feature	Equation	Description	Models
<code>ns(size_mean, df=3)</code>	Natural spline, 3 df	Flexible non-linear size	M21, M23, M25, M34
<code>ns(size_mean, df=4)</code>	Natural spline, 4 df	More flexible size	M22
<code>ns(Weight, df=3)</code>	Natural spline, 3 df	Flexible non-linear weight	M23

12. BOX-COX TRANSFORM (1)

Feature	Equation	Description	Models
<code>Age_bc</code>	<code>(Age^λ - 1) / λ</code>	Optimal power transform	M24

Note: Optimal $\lambda \approx 0$ (equivalent to log transform)

MODEL SUMMARY: Feature Usage

Model	Type	Key Features	MAE
M1	Baseline	Original 8 features	~1.45
M2	Simple	is_infant, size_mean, Weight, shell_thickness_proxy	~1.55
M7	Log	log(Age) + core features	~1.38
M8	Log+Poly	log(Age) + size_mean ²	~1.37
M15	Full Interactions	4 key interactions	~ 1.35
M21	Spline	ns(size_mean, 3)	~1.38
M37	Enhanced Lasso	All interactions + regularization	~1.36
E5	Weighted Ensemble	M15 + M37 + M38	~ 1.34

TOP 10 MOST IMPORTANT FEATURES

1. `size_mean` - Core predictor, appears in all top models
 2. `is_infant` - Critical for distinguishing growth patterns
 3. `shell_thickness_proxy` - Strong age indicator
 4. `shell_ratio` - Shell proportion highly correlated with age
 5. `shucked_ratio` - Meat proportion changes with age
 6. `size_mean_squared` - Captures diminishing growth rate
 7. `weight_infant_interaction` - Infant-specific weight effect
 8. `size_shell_interaction` - Size $\times$ shell proportion
 9. `is_slow_growth` - Flags biological anomalies
 10. `density` - Compactness indicator
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R CODE: Create All Features

```
r
```



```

abalone <- abalone %>%
  mutate(
    # Ratios
    shucked_ratio = `Shucked Weight` / Weight,
    viscera_ratio = `Viscera Weight` / Weight,
    shell_ratio = `Shell Weight` / Weight,
    height_diameter_ratio = Height / Diameter,

    # Volume & Density
    volume_ellipsoid = (4/3) * pi * (Length/2) * (Diameter/2) * (Height/2),
    density = Weight / volume_ellipsoid,
    surface_area = 4 * pi * ((Length * Diameter)^1.6075 / 3)^(1/1.6075),

    # Geometric Means
    size_mean = (Length * Diameter * Height)^(1/3),

    # Shell Thickness
    shell_thickness_proxy = `Shell Weight` / surface_area,

    # Binary Flags
    is_infant = if_else(Sex == "I", 1L, 0L),
    is_slow_growth = if_else(Sex == "I" & Age > 6, 1L, 0L),

    # Polynomials
    size_mean_squared = size_mean^2,
    size_mean_cubed = size_mean^3,

    # Key Interactions
    size_shell_interaction = size_mean * shell_ratio,
    size_shucked_interaction = size_mean * shucked_ratio,
    weight_infant_interaction = Weight * is_infant,
    shell_infant_interaction = shell_ratio * is_infant,

    # Log Transforms
    log_size_mean = log(size_mean),
    log_weight = log(Weight)
  )

```